



End-to-End Multimodal CTR Prediction for Recommendation Systems

Leveraging User Behavior, Interaction Signals and Pretrained Embeddings
with Deep Neural Networks



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1. INPUT DATA



1.1 USER BEHAVIOR

- user_id
- item_seq (user history)
- likes_level
- views_level



1.2 ITEM METADATA

- item_id
- item_tags



1.3 PRETRAINED EMBEDDINGS (Task 1 – Competition Provided)

- 128-dimensional multimodal embeddings for items
- Source: Provided on the competition website



GOAL

Predict the probability that a user will click / engage with a target item.

2. MODEL PIPELINE

2.1 DATA PREPARATION



Map item_id
→ embedding
index

123

Encode categorical
features to
integer format



Construct dense
embedding matrix
(91718 × 128)

2.2 FEATURE REPRESENTATION

User ID
Embedding
(32-dim)



Item ID
Embedding
(32-dim)



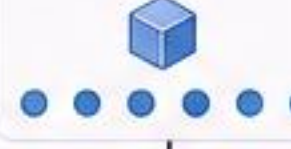
Likes Level
Embedding
(8-dim)



Views Level
Embedding
(8-dim)



Pretrained Item
Embedding
(128-dim)
(Competition
Provided)



CONCATENATION
Total Dimension = 228



2.3 PREDICTION MODEL (DEEP MLP)

Dense (256)
ReLU
BatchNorm
Dropout
(0.3)

Dense (128)
ReLU
BatchNorm
Dropout
(0.2)

Dense (64)
ReLU
Dropout
(0.1)

Output Layer (1)
Sigmoid



CTR Probability
(0 to 1)

LOSS FUNCTION

Binary Cross-Entropy with Logits (BCEWithLogitsLoss)



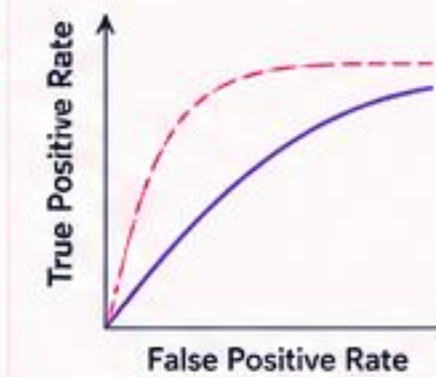
3. OUTPUTS & RESULTS

3.1 PREDICTION OUTPUT



CTR Probability
(0 to 1)
Higher score =
Higher likelihood
of engagement

3.2 MODEL EVALUATION



- AUC Score
(Main metric)
- Train vs Validation
AUC Gap
(Overfitting Analysis)
- Early Stopping &
Model Checkpoint

3.3 FINAL SUBMISSION



prediction.csv
(ID, Task1&2)



prediction.zip
(For submission)

DATA SPLITS



Train
3,600,000
(≈99.2%)



Validation
10,000
(≈0.3%)



Test
379,142
(≈10.5%)



KEY TAKEAWAYS



Combines behavioral signals
(history, likes, views) with
pretrained multimodal item
embeddings (provided in the
competition).



Deep MLP architecture with
regularization (BatchNorm,
Dropout, Early Stopping) for
robust generalization.



Achieves high CTR prediction
performance, improving ranking
and personalization quality in
recommendation systems.



End-to-end pipeline: from raw
data to submission-ready
predictions ensuring
reproducibility and scalability.



FINAL COMPETITION SCORE

0.832

Improvement of +0.057 over baseline



Note: Pretrained multimodal embeddings for items were provided by the competition organizers (Task 1).